SERGEI MOLCHANOV

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Robotics Software Engineer with 2 years of industry experience focused on building ROS2 autonomy stacks for robots, specializing in Navigation and BehaviorTree-based control.

EXPERIENCE

Robotics Software Engineer (Freelance) — Feb 2026 - Present

Future of Life Institute (US Client) - Nonprofit reducing large-scale risks from transformative technologies

- **Flew a 350g quadcopter on fully autonomous, GPS-denied indoor missions** that locate and approach a known face, using custom-patched ArduPilot firmware, optical-flow/LiDAR EKF fusion, and a CUDA-accelerated face-recognition ground station (~60 FPS) commanding the drone over MAVLink
- **Turned an unstable airframe into repeatable end-to-end autonomy** by root-causing instabilities via FFT on raw 1kHz IMU logs, hand-tuning rate PIDs and altitude control, and building a fail-safe mission state machine with verified commands and layered emergency responses
- Developing a Unitree Go2-based project utilizing ROS2 & Nav2

Robotics Software Engineer — Oct 2025 - Feb 2026

Palisade Research (US Client) - AI safety nonprofit researching frontier-model capabilities and controllability

- **Built the autonomy stack for a Unitree Go2 quadruped doing unattended perimeter patrol** as the team's robotics engineer, integrating LiDAR-based SLAM, Nav2, YOLOv8 person detection, and BehaviorTree-based reactive behaviors, and getting non-robotics engineers up to speed on ROS2
- **Sharpened navigation and mapping accuracy** through LiDAR extrinsic calibration and point cloud denoising of the sensor data feeding SLAM and Nav2
- **Ran AI safety experiments on LLM shutdown avoidance** with an LLM controlling the robot, designing the trial protocols and analyzing the behavioral data

Junior Robotics Software Engineer — Jan 2023 - Feb 2024

Coalescent Mobile Robotics (Denmark) - Danish startup building autonomous mobile robots (AMRs) for retail

- **Delivered 15+ mission-specific robot behaviors in C++** on the Nav2 Behavior Tree architecture driving a fleet of AMRs' motion planning
- **Re-architected the Behavior Tree package to orchestrate the entire robotic system (20+ packages)**, raising testability, robustness, and statistics collection
- **Field-tested and debugged the robot on-site** during live deployments

PUBLICATIONS

Shutdown Resistance in Large Language Models, on robots!

Palisade Research Technical Report

Petrov A., Koldyba S., Molchanov S., et al.

EDUCATION

Graduate Studies - Robotics — 2022 - 2023

University of Southern Denmark

- For **Navigation, Guidance & Control**, developed 4-rotor drone rigid body simulation & PID-based stabilization software in Simulink
- For **Embodied AI**, built Mindstorms linefollower with autonomous CV-based grapple

B.Sc. Robotics — 2019 - 2024

ITMO University

- **Thesis**: Comparative evaluation of state-of-the-art object-tracking algorithms
- **Relevant coursework**: C/C++/C# Programming, Control Theory, Algorithms & Data Structures, Machine Learning, Linear Algebra, Linux

SKILLS

Robotics & Autonomy

ROS2 (Humble), Gazebo, Nav2, Behavior Trees, LiDAR, SLAM, Object Tracking, Sensor Integration, ArduPilot (MAVLink)

Programming

C++, Python, C

Systems

Embedded Linux (Jetson, Raspberry Pi), Microcontrollers, Docker

Tools

Git, CMake

COMPETITIONS & AWARDS

Finalist — Jun 2021 - Jul 2022

National CanSat Competition, MSU

- **Led a 3-person team over a year to fly a satellite-in-a-can 1km** and return collected data
- **Wrote low-level ATmega C** linking flight sensors (accelerometer, barometer, etc.), radio, and memory over I2C/SPI/UART, and performed sensor calibration

Hackathon Winner — 2019 & 2020

National Technical Hackathon, Innopolis University

- **Built C++ flight-control and computer-vision software for a drone that autonomously detects and collects colored cubes**, integrating onboard sensors for closed-loop navigation
- **Programmed a rover-like robot to complete a simulated Mars mission** in Python/C++, with remote joystick control, an onboard drill, and a live camera feed

INTERESTS

Acting, Film, Game Design, Yoga

LANGUAGES

English - Proficiency (C2)

Russian - Native

Norwegian - Intermediate (B1)